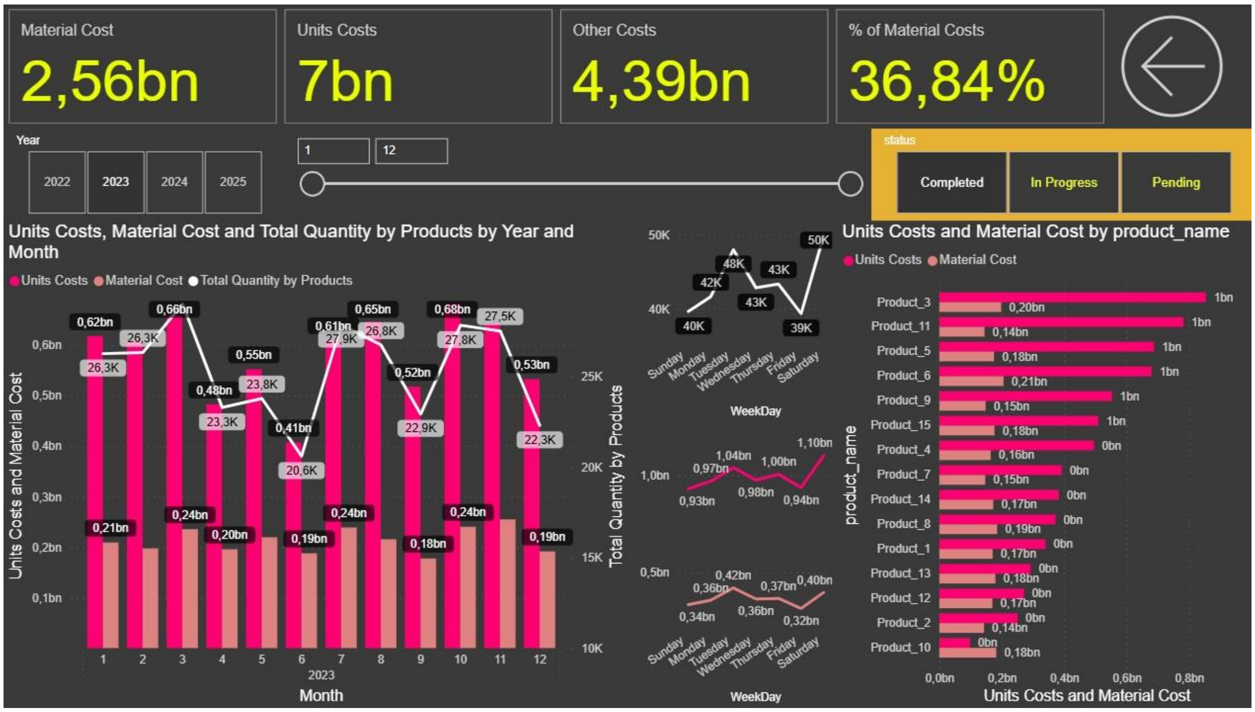
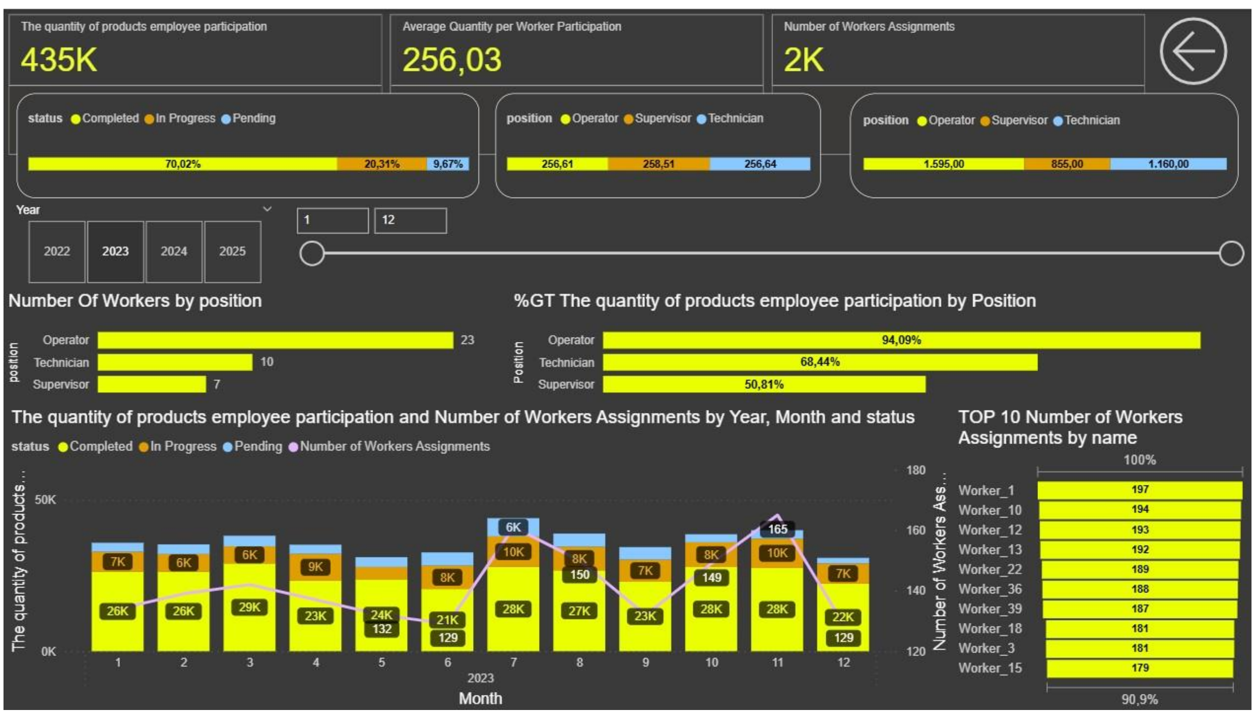
CONCLUSIONS FOR YEAR 2023.




Year
2022
2023
2024
2025
1
12
YoY Change in Production Quantity
1,67%
YoY Change in Unit Costs
2,56%
YoY Change of Material Cost
3,27%
YoY Change of Material Qty
3,18%
YoY Change in Production Quantity by Year and Month
5,05% 1,79% 23,80% -12,91% 21,57% -21,91% 3,60% 3,98% 8,13% -6,60% 5,66% -2,24%
YoY Change in Unit Costs by Year and Month
13,25% 9,77% 33,85% -19,34% 29,35% -31,84% -4,37% 5,83% -2,79% 7,22% 1,86% 2,05%
YoY Change of Material Cost by Year and Month
6,74% -9,50% 14,94% -15,59% 20,65% -2,81% 6,16% 3,60% -1,38% -0,53% 21,58% -0,43%
2023
YoY Change of Material Qty by Year and Month
3,07% -7,30% 11,37% -12,01% 24,02% -3,51% 11,80% -0,28% 1,41% -1,86% 13,01% 3,07%
2023
status
Completed
In Progress
Pending


The quantity of products employee participation
435K
Average Quantity per Worker Participation
256,03
Number of Workers Assignments
2K
status Completed In Progress Pending
70,02% 20,31% 9,67%
position Operator Supervisor Technician
256,61 258,51 256,64
position Operator Supervisor Technician
1.595,00 855,00 1.160,00
Year
2022
2023
2024
2025
1
12
Number Of Workers by position
Operator 23
Technician 10
Supervisor 7
%GT The quantity of products employee participation by Position
Operator 94,09%
Technician 68,44%
Supervisor 50,81%
The quantity of products employee participation and Number of Workers Assignments by Year, Month and status
status Completed In Progress Pending Number of Workers Assignments
The quantity of products...
Number of Workers Ass...
2023
Month
TOP 10 Number of Workers Assignments by name
100%
Worker_1 197
Worker_10 194
Worker_12 193
Worker_13 192
Worker_22 189
Worker_36 188
Worker_39 187
Worker_18 181
Worker_3 181
Worker_15 179
90,9%

🔷 1. Cost Overview

- **Total Unit Costs:** 7bn
- **Material Cost:** 2.56bn → **36.84%** of total unit costs
- **Other Costs:** 4.39bn

✅ **Conclusion:** Material costs are a significant but not dominant part of total unit costs. Non-material (e.g., labor, overhead) costs account for over 60%.

🔷 2. Cost Breakdown by Product

- Products like **Product_3**, **Product_6**, and **Product_5** contribute the most to **unit and material costs**, each nearing or reaching **1bn in unit cost**.
- Some products (e.g., **Product_1**, **Product_7**) show **zero or low material cost**, which could suggest:
  - Service-based items
  - Outsourced production
  - Data quality issues

✅ **Conclusion:** Focus on top 3 products to manage cost impact. Investigate products with zero material cost for anomalies or improvement areas.

🔷 3. Cost Breakdown by Material

- **Top materials by cost:**
  - **Material_14 (0.17bn)**
  - **Material_27 (0.15bn)**
  - Followed by others in the 0.12–0.14bn range.
- **Material Costs per Order** go as high as **1.1M/order**, showing that some materials are extremely expensive per transaction (e.g., **Material_14, Material_3**).

✅ **Conclusion:** Focus on optimizing sourcing for **Material_14** and other high-cost materials. Revisit ordering strategy to reduce per-order expense.

🔷 **4. Time-Based Trends**

📅 **Monthly Breakdown (2023):**

- **Unit Costs and Material Costs** peak in **March, July, and October**.
- **Product quantity** peaked in **March (29K)** and **November (27.8K)**.
- Weekday production is highest on **Saturday (50K)**.

✅ **Conclusion:** Align procurement and labor more efficiently with production peaks. Saturdays might be over-utilized — balance resources better across the week.

🔷 **5. Year-over-Year (YoY) Trends**

✅ **Overall Summary for 2023 YoY Changes:**

- **Production Quantity:** +1.67%
- **Unit Costs:** +2.56%
- **Material Costs:** +3.27%
- **Material Quantity:** +3.18%

➡️ **Insight:** All key indicators increased, but **costs are rising faster than production**, signaling a need to optimize resource usage and cost efficiency.

🔎 **Monthly Analysis and Conclusions:**

January

- 📈 Production: +5.05%
- 📈 Unit Costs: +13.25%
- 📈 Material Costs: +6.74%
- 📈 Material Quantity: +3.07%

✅ **Conclusion:** Significant increase in unit costs—costs are growing faster than output. Investigate cost drivers (materials, labor, overhead).

February

- 📈 Production: +1.79%
- 📈 Unit Costs: +9.77%
- 📉 Material Costs: -9.50%
- 📉 Material Quantity: -7.30%

✅ **Conclusion:** While production slightly increased, material usage and cost dropped. Likely due to improved efficiency or use of cheaper materials.

March

- 📈 Production: +23.80%
- 📈 Unit Costs: +33.85%
- 📈 Material Costs: +14.94%
- 📈 Material Quantity: +11.37%

✅ **Conclusion:** Strong growth across all metrics. Likely a seasonal demand peak. However, costs rising faster than output → potential inefficiencies.

April

- 📉 Production: -12.91%
- 📉 Unit Costs: -19.34%
- 📉 Material Costs: -15.59%
- 📉 Material Quantity: -12.01%

✅ **Conclusion:** A notable drop across the board. Possibly due to downtime, holidays, or decreased demand.

May

- 📈 Production: +21.57%
- 📈 Unit Costs: +29.35%
- 📈 Material Costs: +20.65%
- 📈 Material Quantity: +24.02%

✅ **Conclusion:** Similar to March. Output recovered but costs again outpaced production. Monitor this closely.

June

- 📉 Production: -21.91%
- 📉 Unit Costs: -31.84%
- 📉 Material Costs: -2.81%
- 📉 Material Quantity: -3.51%

✅ **Conclusion:** Sharp decline in production and unit cost. Likely underutilization of capacity or lower activity.

July

- 📈 Production: +3.60%
- 📉 Unit Costs: -4.37%
- 📈 Material Costs: +6.16%
- 📈 Material Quantity: +11.80%

✅ **Conclusion:** Positive productivity trend — higher output with lower unit cost. Efficiency improvement observed.

August

- 📈 Production: +3.98%
- 📉 Unit Costs: -2.79%

- 📈 Material Costs: +3.60%
- 📉 Material Quantity: -0.28%

✅ **Conclusion:** Unit costs are decreasing despite rising material costs, which points to efficiency gains in processes.

September

- 📈 Production: +8.13%
- 📈 Unit Costs: +5.83%
- 📉 Material Costs: -1.38%
- 📈 Material Quantity: +1.41%

✅ **Conclusion:** Higher production achieved with lower material cost — strong operational efficiency.

October

- 📉 Production: -6.60%
- 📈 Unit Costs: +7.22%
- 📉 Material Costs: -0.53%
- 📉 Material Quantity: -1.86%

✅ **Conclusion:** Output dropped while unit costs rose — inefficient month. Possibly due to fixed overheads being spread across fewer units.

November

- 📈 Production: +5.66%
- 📈 Unit Costs: +1.86%
- 📈 Material Costs: +21.58%
- 📈 Material Quantity: +13.01%

✅ **Conclusion:** Material costs spiked disproportionately. Likely due to higher-priced inputs or stockpiling.

December

- 📉 Production: -2.24%
- 📈 Unit Costs: +2.05%
- 📉 Material Costs: -0.43%
- 📈 Material Quantity: +3.07%

✅ **Conclusion:** Output decreased slightly, while costs rose. Material usage increased — may indicate **overstocking** or **inefficient planning**.

📌 **Key Takeaways:**

- **Best-performing months:** July, August, September – improved efficiency, production growth, and lower unit costs.
- **Challenging months:** June, October, December – lower output and rising costs suggest inefficiency.
- **Warning signs:** March, May, and November had high output but even higher cost increases — examine input cost drivers and operational bottlenecks.

🔷 6. Worker Participation

- **435K** total product participation by employees
- **Average contribution per worker:** ~256 units
- **Total assignments:** 2K
- **Operators (23)** are the largest group, contributing **94.09% of participation**.

✅ **Conclusion:** Operators are the workforce backbone — optimize their efficiency. Also, some supervisors show relatively high participation — could indicate role overlap or inefficiency.

🔷 7. Status Distribution

Across different areas (costs, workers, etc.), statuses are distributed as:

- **Completed: 70%**
- **In Progress: 20%**
- **Pending: 10%**

✅ **Conclusion:** Most tasks are on track. Consider strategies to reduce "In Progress" backlog and eliminate unnecessary "Pending" items.

🔷 8. Materials Quantity vs Cost

- Some months (e.g., **August**) show high material cost with **low quantity**, hinting at high-cost materials used.

✅ **Conclusion:** Drill down into such months and materials to control spending or consider alternative suppliers.

🔷 Summary Recommendations:

1. **Optimize top cost-heavy products** (Product_3, Product_6, Product_5).
2. **Negotiate pricing or seek alternatives for expensive materials** (Material_14, Material_3).
3. **Balance weekday production**, especially Saturdays.
4. **Tighten cost control** as material and unit costs rise faster than quantity.
5. **Empower operators** with tools or training as they handle most of the workload.
6. **Improve workflow** to reduce pending/in-progress assignments.

Would you like tailored recommendations based on a specific KPI (e.g., cost reduction, productivity boost, material sourcing)?